

Cauchy point: minimizer of m_k along $-g_k$ (steepest descent)

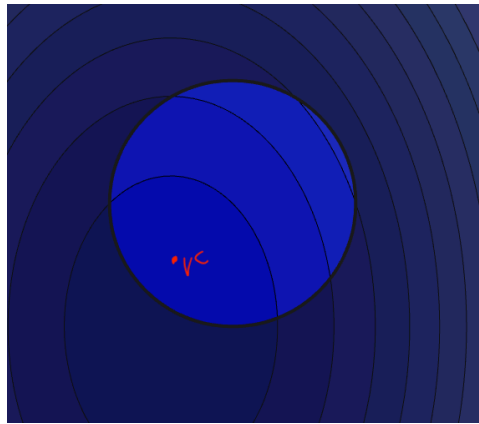
Problem 1:

$$B = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \quad g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \Delta = 1/2$$

Compute an approximation of v^* with Cauchy point method.

curvature: $g^T B g = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 8 > 0 \Rightarrow$ pos. curvature. $\Rightarrow$ find analytic min inside Trust Region.

$$v^C = -t \Delta \frac{g}{\|g\|}, \quad t = \min \left\{ \frac{\|g\|^3}{\Delta g^T B g}, 1 \right\} \Rightarrow \|g\|^2 = \sqrt{2}^2 \approx 2.83 \Rightarrow t = \frac{2.82}{\frac{1}{2}(8)} \approx 0.71$$



$$\Rightarrow \boxed{v^C} = -0.71 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \boxed{\begin{pmatrix} -0.25 \\ -0.25 \end{pmatrix}}$$

$$\|v^C\| = t \Delta \approx 0.35 < 0.5$$

$\Rightarrow$ inside T.R. as expected

Problem 3:

$$B = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \quad g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \Delta = 1$$

Compute the λ^* and v^* which satisfy the KKT optimality conditions.

Case 1: $\lambda_1 = 0$: KKT4) $\begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix} v^* = \begin{pmatrix} -1 \\ -1 \end{pmatrix} \Rightarrow \begin{cases} 4v_1^* = -1 \rightarrow v_1^* = -1/4 \\ 2v_2^* = -1 \rightarrow v_2^* = -1/2 \end{cases}$

$$\|v^*\|^2 = \frac{1}{8} + \frac{1}{4} = \frac{3}{8} < 1 = \Delta^2 \Rightarrow \text{so, } \boxed{v^* = \begin{pmatrix} -1/4 \\ -1/2 \end{pmatrix}, \lambda^* = 0}$$

$\Rightarrow$ within trust region, no need to consider $\lambda_1 > 0$.

KKT Conditions - using $\|v\|^2 \leq \Delta^2$

1.) $\|v^*\|^2 - 1 = 0$

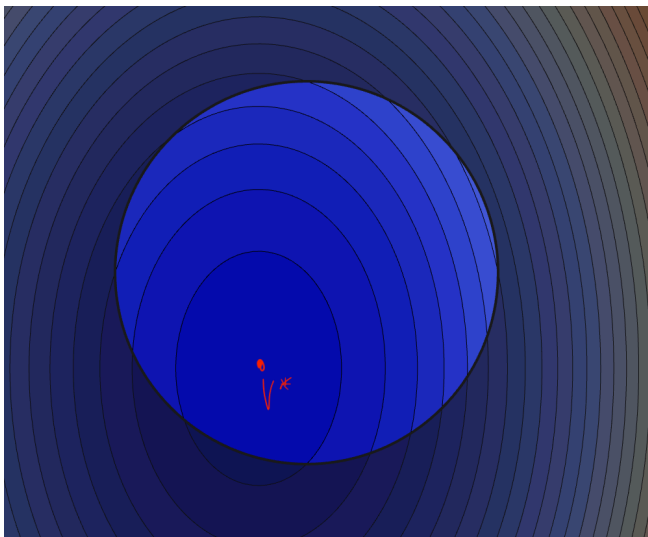
2.) $\lambda_1^* \geq 0$

3.) $\lambda_1^* (\|v^*\|^2 - 1) = 0$

4.) $\mathcal{L}(v, \lambda_1) = f + g^T v + \frac{1}{2} v^T B v + \lambda_1 (\|v\|^2 - 1)$

$$\nabla_v \mathcal{L}(v, \lambda_1) = g + Bv + \lambda_1 v \stackrel{!}{=} 0$$

$$\Rightarrow (B + \lambda_1 I) v^* = -g$$



KKT conditions: using $\|v\|^2 \leq \Delta^2$

Problem 5 (bonus):

$$B = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \quad g = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \quad \Delta = 1/2$$

$$1.) \|v^*\|^2 - \frac{1}{4} \leq 0$$

$$2.) \lambda_1^* \geq 0$$

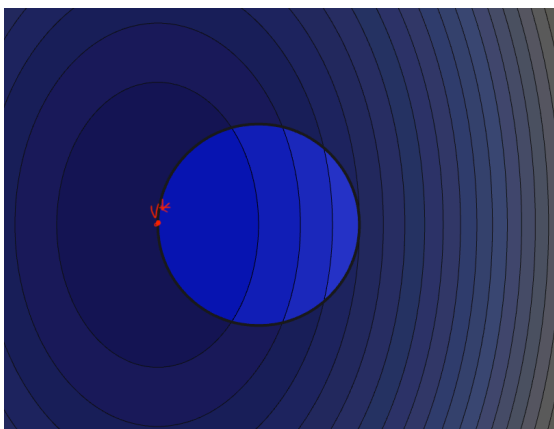
$$3.) \lambda_1^* (\|v^*\|^2 - \frac{1}{4}) = 0$$

$$4.) (B + \lambda_1^* I) v^* = -g$$

Compute the λ^* and v^* which satisfy the KKT optimality conditions.

Case 1: $\lambda^* = 0$ KKT 4: $4v_1^* = -2 \Rightarrow v_1^* = -\frac{1}{2}$
 $2v_2^* = 0 \Rightarrow v_2^* = 0$ $\left\{ \begin{array}{l} \|v^*\|^2 = \frac{1}{4} = \Delta^2 \Rightarrow \text{on boundary} \end{array} \right.$

$$\Rightarrow \lambda^* = 0, \quad v^* = \begin{pmatrix} -\frac{1}{2} \\ 0 \end{pmatrix}$$



Problem 7: (bonus)

Same parameters as in problem 6, but now the gradient is zero $g = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

KKT conditions:

$$1.) \|v^*\|^2 - 1 \leq 0$$

$$2.) \lambda_1^* \geq 0$$

$$3.) \lambda_1^* (\|v^*\|^2 - 1) = 0$$

$$4.) (B + \lambda_1^* I) v^* = -g$$

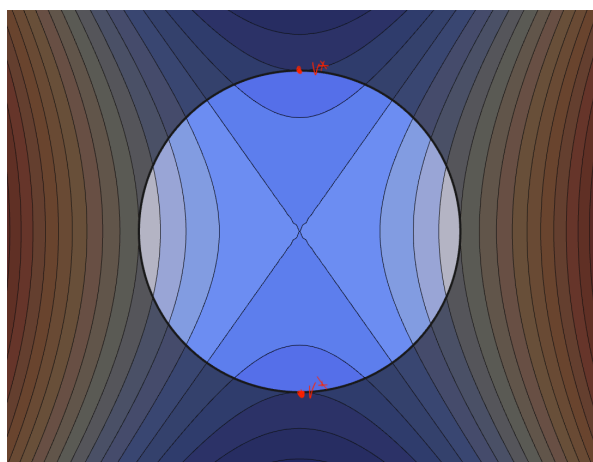
$$5.) B + \lambda_1^* I \in S_+^n \Rightarrow \lambda_1^* \geq 2$$

$$\text{KKT 4.) } (4 + \lambda_1^*) v_1^* = 0 \quad (1)$$

$$(-2 + \lambda_1^*) v_2^* = 0 \quad (2)$$

$$\text{KKT 3.) From KKT 5 we know } \lambda_1^* \geq 2, \text{ so } \|v^*\|^2 - 1 = 0 \Rightarrow v_1^{*2} + v_2^{*2} - 1 = 0 \quad (3)$$

$$\Rightarrow (1), (2), (3): \lambda_1^* = 2, \quad v^* = \begin{pmatrix} 0 \\ \pm 1 \end{pmatrix} \quad (\text{using online solver})$$



Trust Region Subproblem: Exact Solution (1 + Bonus 1 pts)

Consider the following trust region subproblem in two dimensions, $v \in \mathbb{R}^2$:

$$\begin{aligned} &\text{minimize } m(v) = g^T v + \frac{1}{2} v^T B v \\ &\text{subject to: } \|v\| \leq \Delta \end{aligned}$$

as discussed in [Nocedal, Sec 4.3] and the slides. In every problem, given the inputs B, g and Δ , solve the task and plot the quadratic model and the trust region with the **tool**.

Problem 2:

$$B = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \quad g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \Delta = 1/2$$

Compute an approximation of v^* with the Dogleg method.

Let's first see whether the Newton step is inside the trust region. The Newton step is $v^N = -B^{-1}g$. So first I will find B^{-1} using Gauss Jordan

$$[B|I] \rightarrow \left(\begin{array}{cc|cc} 4 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \end{array} \right) \rightarrow \left(\begin{array}{cc|cc} 1 & 0 & 1/4 & 0 \\ 0 & 1 & 0 & 1/2 \end{array} \right)$$

$$\text{So } B^{-1} = \begin{pmatrix} 1/4 & 0 \\ 0 & 1/2 \end{pmatrix} \quad \text{and} \quad v^N = - \begin{pmatrix} 1/4 & 0 \\ 0 & 1/2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1/4 \\ -1/2 \end{pmatrix}.$$

$$\|v^N\| = \sqrt{\frac{1^2}{4} + \frac{1^2}{2}} = \sqrt{\frac{5}{4}} = \frac{\sqrt{5}}{2} = 0.559$$

Newton step is outside the trust region because $\|v^N\| > \Delta = 0.5$

Let's compute $\|g\|$ to know in which dogleg case we are:

$$\|g\| = - \frac{g^T g}{g^T B g} g = - \frac{2}{(1 \ 1) \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = - \frac{1}{3} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1/3 \\ -1/3 \end{pmatrix}$$

$$\|v_g\| = \sqrt{\frac{2}{9}} = \frac{\sqrt{2}}{3} = 0.471.$$

v_g is inside the trust-region so we are in the 3rd case.

$$v^* = v_g + (\gamma - 1)(v^N - v_g) \quad \text{with} \quad \|v^* + (\gamma - 1)(v^N - v_g)\| = \Delta^2$$

$$\begin{aligned} \text{So } v(\gamma) &= \begin{pmatrix} -1/3 \\ -1/3 \end{pmatrix} + (\gamma - 1) \begin{pmatrix} -1/4 + 1/3 \\ -1/2 + 1/3 \end{pmatrix} = \begin{pmatrix} -1/3 \\ -1/3 \end{pmatrix} + \gamma \begin{pmatrix} 1/12 \\ -1/6 \end{pmatrix} - \begin{pmatrix} 1/12 \\ -1/6 \end{pmatrix} \\ &= \begin{pmatrix} -5/12 \\ -1/6 \end{pmatrix} + \gamma \begin{pmatrix} 1/12 \\ -1/6 \end{pmatrix} = \begin{pmatrix} -5/12 + \gamma/12 \\ -1/6 - \gamma/6 \end{pmatrix} \end{aligned}$$

We know that $\|v(\gamma)\|^2 = \Delta^2 \Rightarrow (-5/12 + \gamma/12)^2 + (\frac{1}{6} - \frac{\gamma}{6})^2 = \frac{1}{4} \Rightarrow$

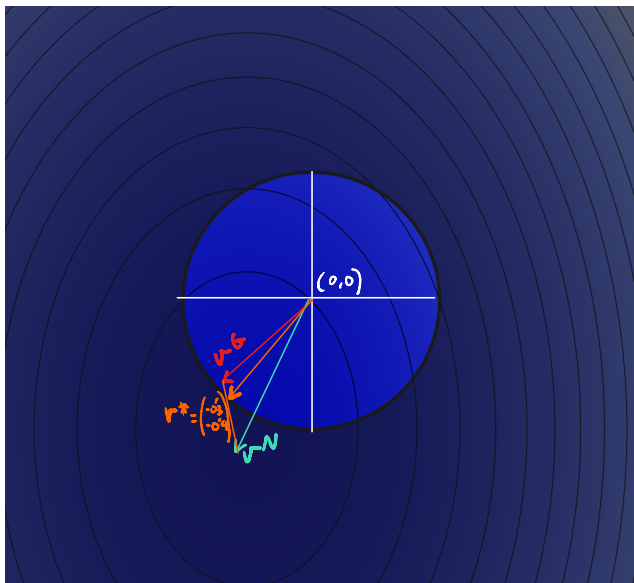
$$\Rightarrow \frac{25}{144} + \frac{-10\gamma}{144} + \frac{\gamma^2}{144} + \frac{1}{36} + \frac{2\gamma}{36} + \frac{\gamma^2}{36} = \frac{1}{4} \Rightarrow$$

$$\Rightarrow \frac{25}{144} - \frac{10\gamma}{144} + \frac{\gamma^2}{144} + \frac{4}{144} + \frac{8\gamma}{144} + \frac{4\gamma^2}{144} - \frac{36}{144} = 0 \Rightarrow$$

$$\Rightarrow 5\gamma^2 - 2\gamma - 7 = 0$$

$$\hookrightarrow \gamma = 1.4$$

$$v(1.4) = \begin{pmatrix} -0.3 \\ -0.4 \end{pmatrix} \quad \|v(1.4)\| = 0.5$$



Problem 4:

$$B = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \quad g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \Delta = 1/2$$

Compute the λ^* and v^* which satisfy the KKT optimality conditions.

KKT Conditions:

$$\mathcal{L}(v, \lambda) = g^T v + \frac{1}{2} v^T B v + \lambda_1 (\|v\|^2 - \frac{1}{4})$$

1.- Primal constraints: $\|v\| \leq 0.5$

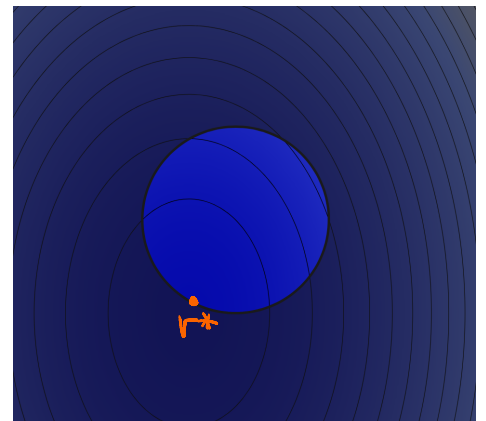
2.- Dual constraints: $\lambda_1 \geq 0$

3.- Complementary slackness: $\lambda(\|v\|^2 - \frac{1}{4}) = 0$

4.- Vanishing gradient of the Lagrangian:

$$\nabla_v \mathcal{L}(v, \lambda) = g + Bv + \lambda_1 v = 0$$

$$(B + \lambda_1 I)v = -g$$



Then for condition 4. to be true:

$$v(\lambda) = \begin{pmatrix} -\frac{1}{4+\lambda} \\ -\frac{1}{2+\lambda} \end{pmatrix}$$

As we know that the solution for the unconstrained problem lies outside the boundary, by complementary slackness v^* is in the boundary.

$$\|v(\lambda^*)\|^2 = \frac{1}{4} \Rightarrow \frac{1}{(4+\lambda^*)^2} + \frac{1}{(2+\lambda^*)^2} = \frac{1}{4} \quad \left. \begin{array}{l} \text{using an online solver:} \\ \lambda^* = 0.264 \end{array} \right\}$$

$$\text{Then } v^* = v(\lambda^*) = \begin{pmatrix} -0.234 \\ -0.446 \end{pmatrix}$$

Problem 6: (bonus)

Consider now the the indefinite matrix

$$\tilde{B} = \begin{pmatrix} 4 & 0 \\ 0 & -2 \end{pmatrix}, \quad g = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \Delta = 1$$

Compute the λ^* and v^* which satisfy the KKT optimality conditions.

KKT Conditions:

$$L(v, \lambda) = g^T v + \frac{1}{2} v^T B v + \lambda_1 (\|v\|^2 - 1)$$

$$1.- \text{Primal constraints: } \|v\| \leq 1$$

$$2.- \text{Dual constraints: } \lambda_1 \geq 0$$

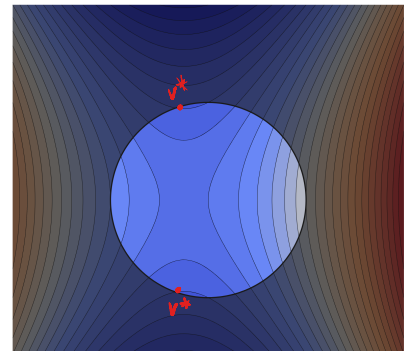
$$3.- \text{Complementary slackness: } \lambda_1 (\|v\|^2 - 1) = 0$$

$$4.- \text{Vanishing gradient of the Lagrangian:}$$

$$\nabla_v L(v, \lambda_1) = g + Bv + \lambda_1 v = 0$$

$$(B + \lambda_1 I)v = -g$$

$$5.- B + \lambda_1 I \in S_+^n \Rightarrow \lambda_1^* \geq 2$$



From KKT4

$$(4 + \lambda_1^*)v_1^* = -2 \quad (1)$$

$$(-2 + \lambda_1^*)v_2^* = 0 \quad (2)$$

From KKT3

$$\lambda_1^* \geq 2 \Rightarrow \|v^*\|^2 - 1 = 0 \Rightarrow (v_1^*)^2 + (v_2^*)^2 - 1 = 0 \quad (3)$$

Using online solver with equations (1), (2) and (3):

$$v^* = \begin{pmatrix} -1/3 \\ \pm \frac{2\sqrt{2}}{3} \end{pmatrix} \quad \lambda_1^* = 2$$